# Simulation of the “Place Charles de Gaulle” Roundabout (Arc de Triomphe, Paris, France)

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**Abstract**

The Place Charles de Gaulle roundabout is highly complex due to the concurrence of multiple approaches/exits, heterogeneous flows, and priority rules. This work models and simulates in AnyLogic the operation of the ring around the Arc de Triomphe, with exit choices weighted by the number of lanes (weights 3–2–1) and traffic-signal control for opposing approach pairs (180s cycle: 30s green / 150s red). The workflow integrates 2D and 3D representations of the aforementioned roundabout, vehicle arrival generation based on access type, and routing with probabilistic destination assignment. Furthermore, it measures the most critical metric aimed for reduction in this study: the average time a vehicle spends in the system, from arrival at any of the 12 entry points to departure via any of the 12 exit points. This case study examines traffic flow during peak hours, with vehicle arrival rates established accordingly. These arrival rates differ based on the number of lanes per street; a higher number of lanes corresponds to a higher arrival rate.

**Keywords**

Simulation - Arc de Triomphe - Roundabout - Traffic Congestion - Peak Hours

## 1. Introduction

In recent years, the evolution of computational power and modelling tools has enabled the operational representation of complex road systems under various assumptions and levels of detail. In this work, we leverage this advancement to study, through simulation in AnyLogic, the performance of the *Place Charles de Gaulle* roundabout (Arc de Triomphe, Paris), characterized by multiple access and exit points, heterogeneous flows, and priority rules that give rise to complex interactions.

In this context, the combination of **Discrete Event Simulation** with **Agent-Based Simulation** allows for capturing driving decisions at a micro level (entry, lane changes, bus stops) and observing their macro effects on the *average time vehicles spend in the system*, both overall, and for cars and buses separately. Furthermore, the **2D/3D geometric modelling** and the **probabilistic destination assignment** —weighted by lane count (weights 3–2–1)— provide a consistent representation of the relative capacity of each exit road. The combination of these techniques facilitates experimenting with demand and control configurations (traffic-signal phases applied to opposite pairs, a 180-sec cycle with 30 sec green and 150 sec red) under peak hour conditions.

The integration of both approaches —microsimulation for internal dynamics and exit weighting to reflect capacity based on lane count and, consequently, vehicle space— enables a precise evaluation of the vehicle’s **average time spent in the system**, *measured from arrival at one of the* 12 *access roads until departure via one of the* 12 *exit roads*. This constitutes the central metric we aim to reduce through scenario analysis.

It should be noted that, in the present study, the scope of the model is deliberately limited. Pedestrian movement and the representation of the second outer roundabout are not included; buses are modelled purely as vehicles, without considering passenger dynamics (boarding and alighting). A homogeneous vehicle fleet is assumed, along with arrival rates differentiated by lane count (the higher the number of lanes, the higher the rate). This methodological decision addresses the need to approach the problem with a clear and replicable formulation, focused on structural elements: geometry and lanes, priority rules, demand per access, and exit distribution. On this basis, various scenarios are explored with the objective of providing quantitative evidence for the management and improvement of system performance, that is, better circulation through the Place Charles de Gaulle roundabout surrounding the Arc de Triomphe in Paris.

### 1.1. Problem Statement

The Place Charles de Gaulle roundabout in Paris represents one of the most iconic and congested traffic hubs in the world. Its circular structure connects twelve main avenues:

1. Avenue des Champs-Élysées
2. Avenue de la Grande-Armée
3. Avenue Kléber
4. Avenue Victor-Hugo
5. Avenue Foch
6. Avenue Hoche
7. Avenue Carnot
8. Avenue Mac-Mahon
9. Avenue d'Iéna
10. Avenue Marceau
11. Avenue de Friedland
12. Avenue de Wagram

As can be observed, among them is the "Avenue des Champs-Élysées", one of the city's most emblematic avenues, generating complex traffic dynamics characterized by constant flows and multiple simultaneous interactions.

Furthermore, the roundabout lacks internal lane markings; however, an analysis of current videos revealed that it has space for approximately eight vehicles in parallel, i.e., eight lanes.

This configuration frequently causes bottlenecks, sudden stops, and minor accidents due to the lack of clear lane markings and the high density of vehicles converging during peak hours.

When modelling the system within the simulation software (AnyLogic in our case) and running initial tests, it was observed that the system saturates rapidly.

Therefore, we deduce that using real data for arrival rates causes frequent saturation due to the highly structured behaviour of agents (cars and buses) in the simulated model. In this regard, observing videos of peak hours at the roundabout reveals erratic and seemingly illogical behaviours that cannot be replicated in the simulation tool. For example:

- Cars stopped at 90° to the normal flow direction, sometimes even facing oncoming traffic:

- Vehicles crossing multiple supposed lanes at once, sometimes more than one vehicle at a time:

Despite its historical and urban value, the roundabout presents serious operational difficulties from a traffic perspective. The absence of internal traffic lights and the coexistence of different vehicle types —private cars, buses, and motorcycles— aggravate flow problems, affecting both traffic efficiency and road safety.

These conditions make the *Place Charles de Gaulle* an ideal case for analysis via computational simulation, as it allows for precise modelling of vehicular flow behaviour and the evaluation of improvement alternatives without physical intervention in the real environment.

During the simulation, vehicle arrival rates were uniformly attenuated using a variable named `lambdaScale`. This variable allows for controlling the system's load level and analysing how the roundabout responds to different congestion scenarios. For example, if the simulation flows without saturation with a `lambdaScale` value equal to 0.3 (rates reduced to 30 %), and after implementing a change the system begins to saturate at the same value, it is considered that the modification worsens performance and is not recommended. Conversely, if saturation occurs only at higher values, for example with `lambdaScale` = 0.5 (rates at 50 %), it can be deduced that the effective utilization of the roundabout increases by 20 %, indicating an improvement in its operational capacity.

### 1.2. Problem

How can vehicular flow at the Place Charles de Gaulle roundabout be optimized, reducing congestion levels and waiting times, through the simulation of various traffic control scenarios in AnyLogic?

### 1.3. General Objective

To analyse and optimize vehicular flow at the Place Charles de Gaulle roundabout through simulation in AnyLogic, by identifying factors leading to congestion and evaluating improvement strategies aimed at enhancing traffic fluidity and safety.

### 1.4. Specific Objectives

- To model vehicular traffic behaviour at the Place Charles de Gaulle roundabout in AnyLogic.
- To identify critical points and primary causes of congestion at the various access points.
- To simulate alternative scenarios using different control strategies by implementing variations in the traffic-signal control system.
- To evaluate the impact of each scenario on the average time spent within the zone and the vehicular density of the roundabout.
- To propose optimization measures that enhance flow efficiency and reduce collision risks.

## 2. Data Collection

Regarding the collection of real data to be implemented in the simulation, an estimation was conducted using a Parisian hourly traffic *dataset*, with the objective of obtaining data corresponding to peak hours (between 17:00 and 20:00). First, street names were normalized (converted to lowercase, accents removed, and hyphens/underscores replaced), and records corresponding to the twelve avenues converging at the *Place Charles de Gaulle* roundabout were filtered.

For those avenues whose names appeared abbreviated or with different aliases, such as `Av_Kleber_Trocadero`, `Hoche-Tilsitt`, or `Carnot-Tilsitt`, search patterns were expanded to guarantee their correct identification.

Based on the *Débit horaire* field (vehicles/hour), the hourly average per avenue was calculated, and data for the 17:00 to 20:00 time frame was filtered. Using these values, the Poisson distribution parameter $\lambda$ was obtained through the following formulas:

$$\lambda_{\text{minute}} = \frac{\text{Average Hourly Flow}}{60}$$

$$\lambda_{\text{second}} = \frac{\text{Average Hourly Flow}}{3600}$$

The result expresses the average car arrival rate (cars/minute and cars/second) for each avenue within that time slot.

The Poisson model is used because, under normal urban traffic conditions with traffic signals, car arrivals can be considered a random process where they arrive independently and, over short intervals (within each time slot), with an approximately constant average rate. The presence of traffic signals generates small batches or *"platoons"* of cars, but when combined with different flows originating from various avenues and staggered signal cycles, the resulting process can be modelled as a Non-Homogeneous Poisson Process (NHPP), with a rate $\lambda(t)$ varying between time slots.

In simulation practice, this NHPP is approximated by a set of stationary Poisson processes, one for each time slot, using its corresponding $\lambda$ value. In terms of implementation, this value is applied in AnyLogic's `Source` block by configuring the arrival type as *Random (Poisson)* and entering the estimated $\lambda$ value (cars/minute). The calculated values were as follows:

| Arrival rate (λ) [number of cars/min.] | | | | | |
|---|---|---|---|---|---|
| Streets | 100% | 80% | 60% | 40% | 20% |
| 1 | 11 | 8,8 | 6,6 | 4,4 | 2,2 |
| 2 | 15 | 12 | 9 | 6 | 3 |
| 3 | 12 | 9,6 | 7,2 | 4,8 | 2,4 |
| 4 | 5 | 4 | 3 | 2 | 1 |
| 5 | 11 | 8,8 | 6,6 | 4,4 | 2,2 |
| 6 | 7 | 5,6 | 4,2 | 2,8 | 1,4 |
| 7 | 8 | 6,4 | 4,8 | 3,2 | 1,6 |
| 8 | 9 | 7,2 | 5,4 | 3,6 | 1,8 |
| 9 | 6 | 4,8 | 3,6 | 2,4 | 1,2 |
| 10 | 13 | 10,4 | 7,8 | 5,2 | 2,6 |
| 11 | 7 | 5,6 | 4,2 | 2,8 | 1,4 |
| 12 | 13 | 10,4 | 7,8 | 5,2 | 2,6 |

The calculation of arrival rate reduction percentages is included as they will be useful later for conducting comparisons. Each index corresponds to the number assigned to each avenue in the list above.

## 3. AnyLogic Modelling Methodology

This section describes the workflow followed to build, implement, and experiment with the model in AnyLogic, from geometric representation to metric collection and scenario comparison.

### 3.1. Geometric Construction and Libraries

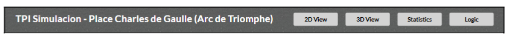
The **Road Traffic Library** was used to define the road infrastructure:

- **Ring and spokes**: creation of the roundabout ring with eight lanes and 12 spokes (access roads and exit roads) with 1, 2, or 3 lanes depending on the case.
- **Nodes and connections**: straight/curved sections connected to entry/exit nodes; ring radii adjusted to avoid overlaps and impossible turns.

### 3.2. Graphical Interface and Navigation

In addition to the geometric construction and internal model logic, a graphical interface was designed to facilitate exploration of the simulated environment. The main presentation and navigation elements incorporated are described below:

- **2D/3D Presentation**: layout of the *3D Window* for better visualization of the environment and configuration of the experiment's *viewArea* to start the simulation directly in 3D.
- **Navigation Menu**: a series of buttons is offered at the top, facilitating navigation through the model (2D/3D visualization, statistics section, logic section).



### 3.3. Agents and Operational Parameters

The model utilizes **multiple vehicle types** as independent agents to facilitate visual inspection, differentiating cars with distinct colours and, where applicable, specific parametrizations by type:

- **`Car` Agents**: various car classes were defined (`BlackCar`, `WhiteCar`, `RedCar`, `GreenCar`, `DarkGreenCar`, `MaroonCar`, `PurpleCar`), which inherit from the standard vehicle of the *Road Traffic Library*. These classes share the same default dynamic parameters (length, acceleration, speed, etc.) and the same routing logic; they differ only in their *appearance* (colour), facilitating the visual inspection of trajectories in the simulation.

- **`Bus` Agent**: represents larger transport vehicles and is also based on the corresponding class in the *Road Traffic Library*. Its geometric and dynamic characteristics (e.g., greater length and different acceleration than cars) are those provided by the library by default; therefore, it behaves in a more conservative manner when circulating through the roundabout, without specific additional logic incorporated into this agent. Distinct arrival rates were established compared to those for cars, without differentiating in this case based on the access road used by the bus.

The system's **operational parameters** are:

- **Traffic-signal control:** 180 sec. cycle with 30 sec. green and 150 sec. red for opposite pairs. The configuration of the `Traffic Light` blocks from the `Road Traffic Library` synchronizes signal groups (pairs of accesses sharing the same phase).

- **Priority rule:** priority is assumed for the *access roads* over the roundabout's internal circulation since, according to gathered information, this is how the circulation is designed in this roundabout in reality.

- **Demand per access (peak hour):** arrival rates differentiated according to lane count.

- **Vehicle attributes:** each agent (distinct `Car` and `Bus`) is assigned a *destination* (exit established by 3–2–1 distribution), and stores an *arrival timestamp* and an *exit timestamp*, to calculate the *average time spent in the system* as the simulation progresses (both for cars and buses separately, as well as a weighted average of both combined). This information is displayed in *three charts*, one for each distinct metric:


The weighted average between cars and buses (first chart) is calculated using the following formula:

$$\bar{t} = \frac{\bar{t}_{\text{car}}\, n_{\text{car}} + \bar{t}_{\text{bus}}\, n_{\text{bus}}}{n_{\text{car}} + n_{\text{bus}}}$$

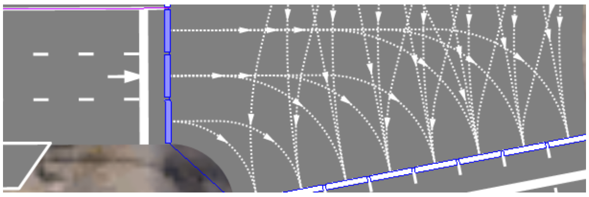
where

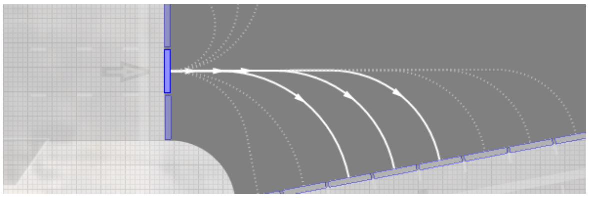
- $\bar{t}_{\text{car}} = \text{mean}\big(\texttt{carArrival}\big)$
- $\bar{t}_{\text{bus}} = \text{mean}\big(\texttt{busArrival}\big)$
- $n_{\text{car}} = \text{count}\big(\texttt{carArrival}\big)$
- $n_{\text{bus}} = \text{count}\big(\texttt{busArrival}\big)$

### 3.4. Arrival Generation and Internal Routing

Arrivals are modelled using the `CarSource` block per access and vehicle type (car and bus), parametrized with the previously mentioned rates.

Please refer to the following images:





As evidenced, we restricted the set of lanes through which a vehicle can enter the roundabout, depending on the lane it is currently circulating in on the access road. The case presented in the images corresponds to the Avenue de la Grande Armée, which has 3 lanes entering the roundabout. Thus, it was determined that if a vehicle is travelling in the lower lane shown in the image (the right lane of the access road), it can only enter the roundabout through one of its two outer lanes. If the vehicle is travelling in the centre lane, it can enter through one of the next three lanes (more interior than the previous ones; this is the specific case shown in the second image). Finally, if the vehicle is travelling in the left lane, it can only enter the roundabout through one of its 3 innermost lanes.

This decision —not to permit a vehicle approaching from any lane on the access road to enter the roundabout through any of its 8 lanes randomly— was made to mitigate further bottlenecks. Permitting unrestricted access led to situations where, for example, a vehicle approaching in the centre lane of the access road wanted to enter an outer lane of the roundabout, while another vehicle approaching in the right lane of the access road sought to enter a more interior lane. The consequence was mutual obstruction; the vehicle approaching in the centre lane experienced significant delays, waiting through multiple signal cycles to enter, as it yielded to all vehicles approaching in the right lane. This significantly increased vehicle queuing on the access road, leading to saturation in certain cases as new vehicles continued to arrive.

The methodology described above was implemented for each of the 12 access roads, segmenting entry into the roundabout's 8 lanes based on the number of lanes available on the access road. For instance, on access roads with 2 lanes, vehicles from each lane can enter 4 specific lanes of the roundabout. On access roads with a single lane, vehicles may enter any of the roundabout's 8 lanes.

### 3.5. Probabilistic Exit Assignment (3–2–1 Weighting)

Each vehicle receives a **destination** according to a distribution proportional to the lane count of each exit. Let $w_j \in \{3,2,1\}$ be the weight of exit $j$ and $W = \sum_j w_j$ be the sum of weights; the probability is:

$$p_j = \frac{w_j}{W}.$$

In the studied case (2 exits with 3 lanes, 6 with 2 lanes, and 4 with 1 lane), we have
$W = 2\times 3 + 6\times 2 + 4\times 1 = 22$.

To define the probability of a vehicle exiting via a specific avenue, a method was used that assigns a higher probability to those possessing more exit lanes. The method is simple yet functional for our case. It consists of dividing 1 (the total sum of probabilities) by the total number of possible exit lanes, which in this case is 22:

$$\text{ProbExitPerLane} = \frac{1}{22} = 0{,}0454545$$

Based on this base value, each street is assigned a probability proportional to the number of exit lanes it has:

$$\text{ProbExitAv3Lanes} = 3 \times \text{ProbExitPerLane} = 0{,}1364$$

$$\text{ProbExitAv2Lanes} = 2 \times \text{ProbExitPerLane} = 0{,}0909$$

$$\text{ProbExitAv1Lane} = 1 \times \text{ProbExitPerLane} = 0{,}04545$$

In total, there are two streets with three exit lanes, six streets with two lanes, and four streets with a single lane. Verifying the sum of all probabilities results in:

$$2 \times 0{,}1364 + 6 \times 0{,}0909 + 4 \times 0{,}04545 = 1$$

which confirms that the probability distribution is consistent.

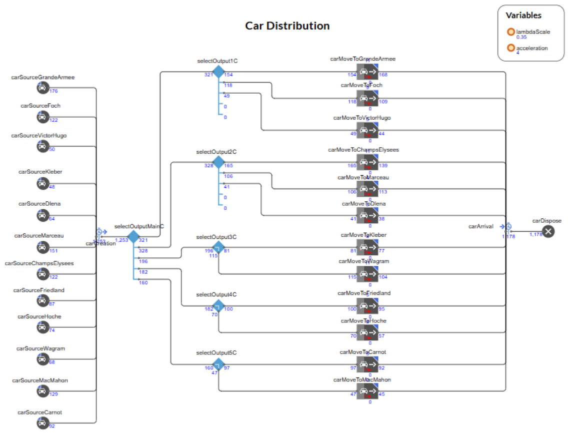
The implementation was carried out using a `SelectOutput5` **tree** (due to the limit imposed by AnyLogic of 5 branches per block), chaining other `SelectOutput5` and `SelectOutput` blocks, from which the distinct `CarMoveTo` blocks originate until covering all 12 exits, while maintaining the global probabilities.

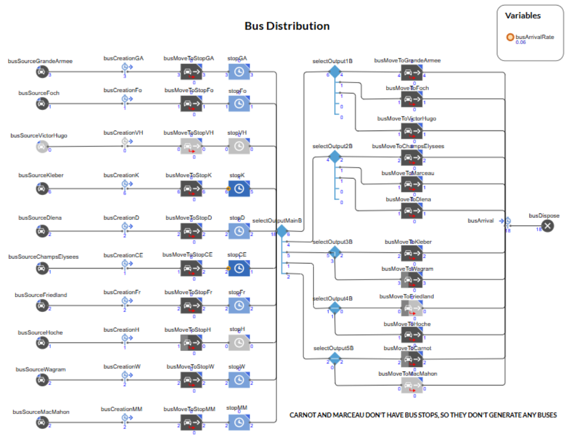
Finally, a `CarDispose` block was introduced so that vehicles are removed from the system once they reach the end of their probabilistically selected exit road.

Furthermore, prior to the first `SelectOutput5`, we utilized a `TimeMeasureStart` block, tasked with initiating timing for each vehicle as it appears at the beginning of any of the access roads. Linked to this timer, we utilized a `TimeMeasureEnd` block before the `CarDispose`, which terminates the previously initiated timing.

This allows us to measure the time taken by each car from its appearance on one of the 12 access roads until it leaves the system via one of the 12 exit roads. With this metric measured for each vehicle, we can subsequently calculate the **average time spent in the system** at the end of the simulation.

Below, we present two images corresponding to the described tree diagram. The first corresponds to the distribution for cars, while the second corresponds to that for buses:



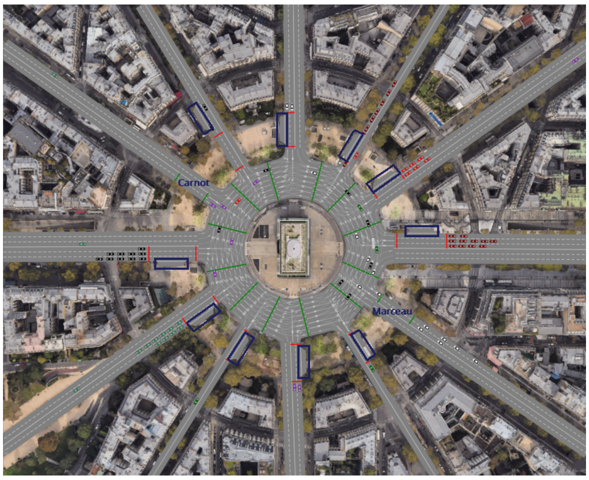

Both diagrams coincide from the probabilistic decision tree onwards, differing in the interval from when the bus is generated until its destination is decided by said tree.

Unlike cars, buses make a stop just before entering the roundabout, simulating their actual behaviour. Due to this, we utilize a `CarMoveTo` block that directs them to the corresponding stop on the access road through which they are entering. Once they reach it, the `Delay` block from the `Process Modelling Library` establishes a stop time for the bus before it continues its journey and enters the roundabout. This block is assigned a waiting time governed by a `triangular(8, 12, 15)` distribution, implying it stops between 8 and 15 seconds, with a mean/mode value of 12 seconds, representing the most fre-

quent value.
Once the delay/stop concludes, the bus continues its path, and through the decision tree explained in detail previously, an exit road is defined for it, towards which it proceeds by circulating through the roundabout once the access road traffic light permits its entry.

### 3.6. Introduction of Buses into the System

As already mentioned in previous sections of this report, we introduced buses into the simulation model. This decision was taken with the objective of adding greater complexity and making the model more closely resemble reality. To do this, we utilized a `Bus Agent`, which represents buses entering the system. To complement this, we added bus stops only on the streets that actually have them. This implied adding them to 10 of the 12 access roads, with the exception of Avenue Carnot and Avenue Marceau, which happen to be located opposite each other.




We defined a lower arrival rate for buses compared to cars, taking into account the actual frequency with which they usually arrive. We established this rate at a value of **0.06 buses per minute** for each of the 10 available access roads, which is equivalent to **one bus approximately every 16 minutes and 40 seconds**. This rate was attributed to the 10 access roads equally, without differentiating based on their lane count, to simplify the calculation.

The arrival rate is clearly lower than those at which cars arrive, where they range between **1.75 and 5.25 cars per minute**. The difference between car arrival rates is due to the fact that more of them arrive via access roads that have more lanes. These numbers mean that cars will arrive via the 12 access roads with a frequency of between **approximately 11 and 34 seconds**.
As mentioned in Section 2, car arrival rates were obtained through the collection of real traffic data in the zone. These rates were attenuated with a scaling factor of 0,35 to facilitate the development and execution of the simulation, since in reality vehicles take unusual manoeuvrers that cannot be represented in the simulation (vehicles crossing at 90° between lanes, dangerous lane crossings, etc.).
See the images in Subsection 1.1.

### 3.7. Traffic-Signal Control

Three variants based on the traffic-signal control system were modelled, whose signals can be seen in the image in the previous section (red and green lines). The three alternative systems proposed for analysis were:

- Baseline system (real) with a single traffic-signal per access road, located next to the pedestrian crossing between the main and outer roundabouts
- Double traffic-signal system per access road, synchronized with each other
- Staggered double traffic-signal system per access road, in which the signal furthest from the main roundabout turns green first, and 5 seconds later, the one located almost at the roundabout entry does so

In all three cases, active signals remain green for 30 seconds, and those on opposite access roads are synchronized, meaning only vehicles arriving via those two roads enter the roundabout simultaneously.
Furthermore, the activation of signals by pairs of opposite roads is carried out in a ring pattern, starting with the signals of the main 3-lane avenues (Avenue des Champs-Élysées and Avenue de la Grande Armée) turning green. At the moment these signals turn red, the signals of the road to the right of each one turn green. In this way, the green signals rotate in a ring formation.
In the staggered signal system, the same applies, except that the ring rotation is synchronized among signals at the same position. That is, the signals furthest from the main roundabout rotate synchronously in a ring pattern, and separately, the signals located at the roundabout entry do the same.
Since signals remain green for 30 seconds, and there are 6 pairs of opposite roads synchronized with each other, each signal must remain red for $30 * (6-1) = 150$ seconds, aiming to maintain the premise that vehicles from only two roads enter the roundabout at a time. Thus, each signal waits for all the green cycles of the rest of the roads in the ring to complete until its turn comes around again.

### 3.8. Run Configuration

To ensure comparability, each **alternative traffic-signal control system** (baseline, synchronized double signal, and staggered double signal) was evaluated using the **same seeds** $\{1,2,3,4\}$. In replication $k$ of each system, all other parameters remained unchanged (car and bus arrival rates, vehicle acceleration, exit probabilities across the 12 roads, etc.). Thus, any observed difference between results was attributed *exclusively* to the **traffic-signal control strategy**.

### 3.9. Three-Dimensional (3D) Visualization

As we explained in previous sections, in addition to the 2D Visualization shown in the image in Section 3.6, we implemented a 3D Visualization using the three-dimensional appearance of the `Car` (in different colors) and `Bus` Agents, and adding custom 3D models to represent the *Arc de Triomphe*, *bus stops with seats*, and *trees*.
The following image shows the aforementioned 3D Model:

### 3.10. AnyLogic Cloud Model

The developed model is available on the AnyLogic Cloud and is publicly accessible at the following link:

https://cloud.anylogic.com/model/b8b776ec-7e67-457b-bcc7-30be0c6e58a6

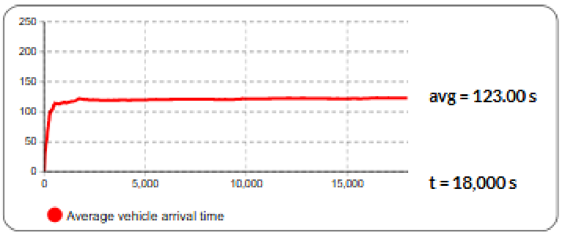
## 4. Results

To evaluate model performance, four runs of 300 simulated minutes were executed for each alternative system, varying only the traffic-signal control strategy. In each simulation, the average time cars take from entry via any of the 12 access roads to exit via any of the 12 egress roads was measured, along with the average time for buses and a weighted average time combining both vehicle types, calculated as described in Section 3.3.
The results obtained from these runs are presented below:

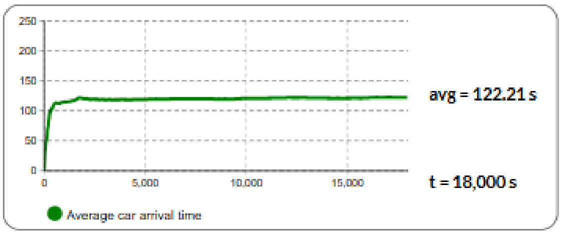
### 4.1. Baseline system (real) with one signal per access, next to the outer ring

**Seed** fixed at a value of **1**:

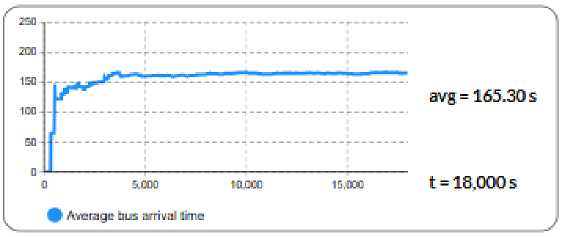
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

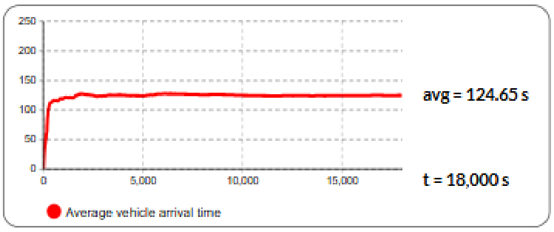



Average bus arrival time (seconds)

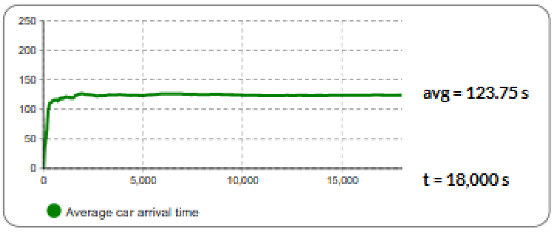



**Seed** fixed at a value of **2**:

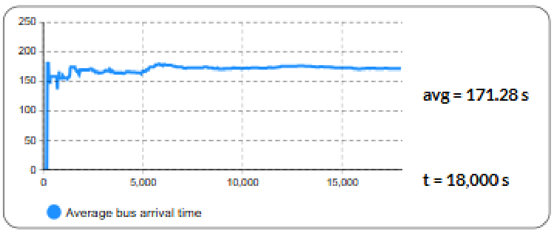
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

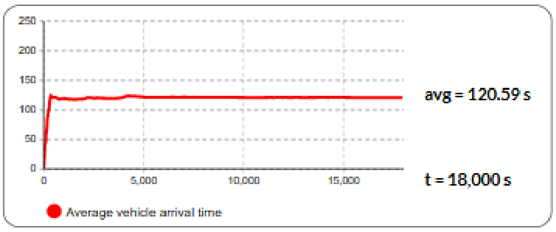



Average bus arrival time (seconds)

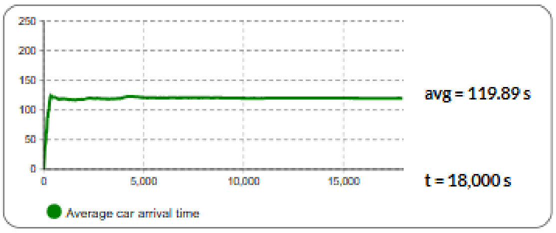



**Seed** fixed at a value of **3**:

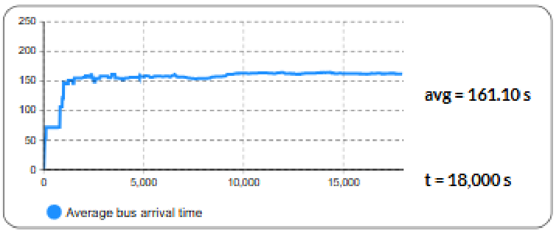
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

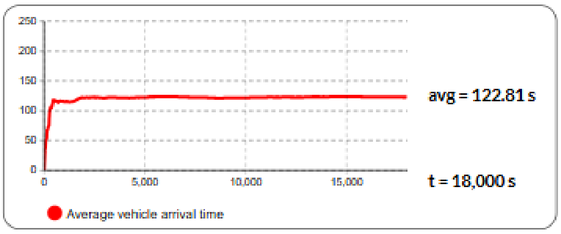



Average bus arrival time (seconds)

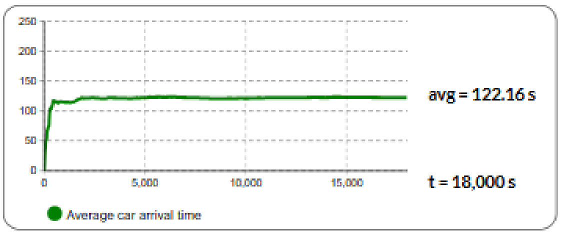



**Seed** fixed at a value of **4**:

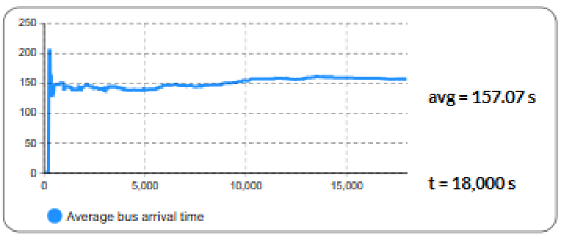
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

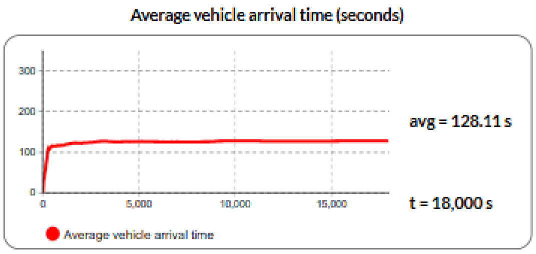


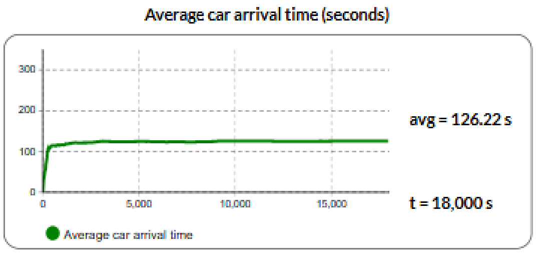

Average bus arrival time (seconds)

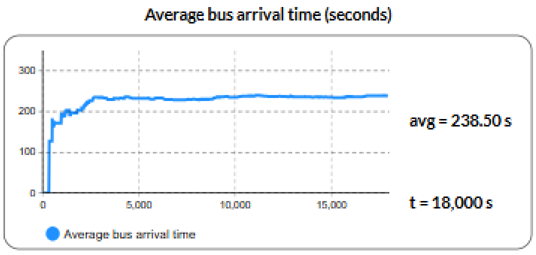


## 4.2. Synchronized double traffic-signal system per access road

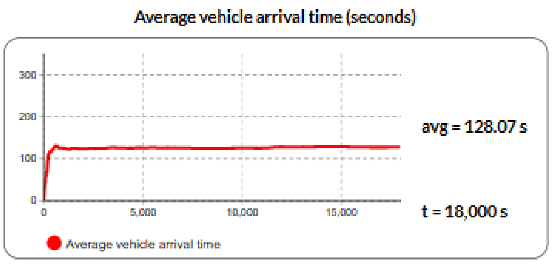
**Seed** fixed at a value of **1**:

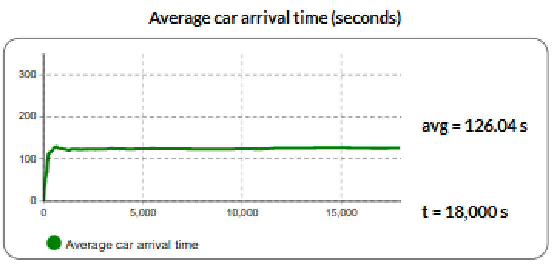


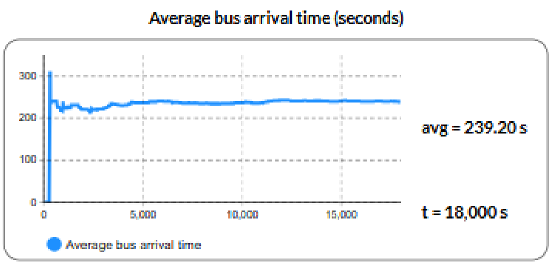



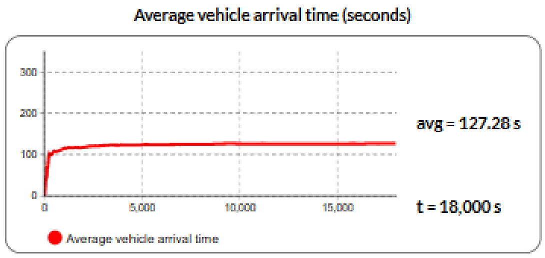

**Seed** fixed at a value of **2**:

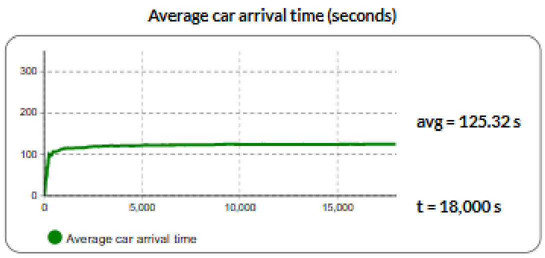


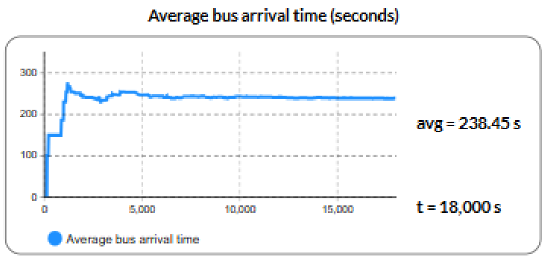



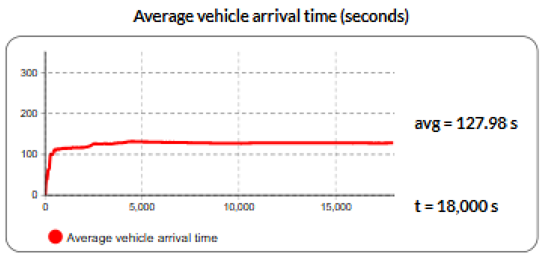

**Seed** fixed at a value of **3**:

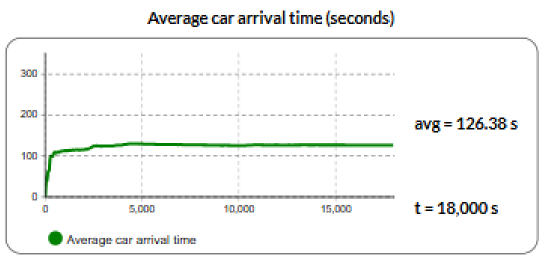


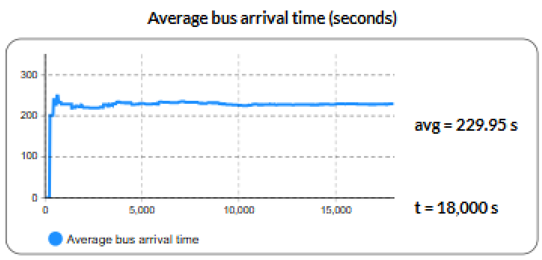




**Seed** fixed at a value of **4**:



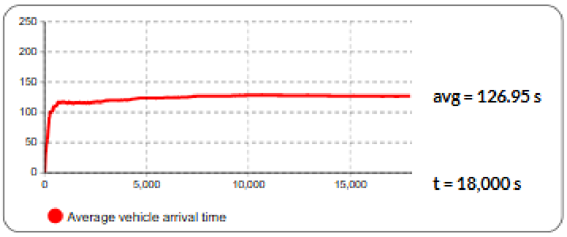



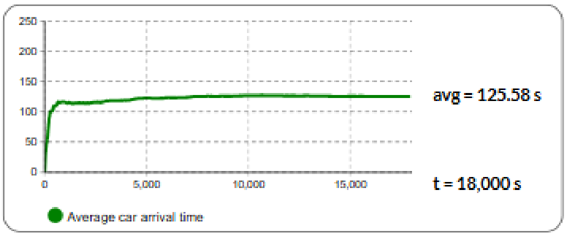
### 4.3. Staggered double traffic-signal system per access road

**Seed** fixed at a value of **1**:

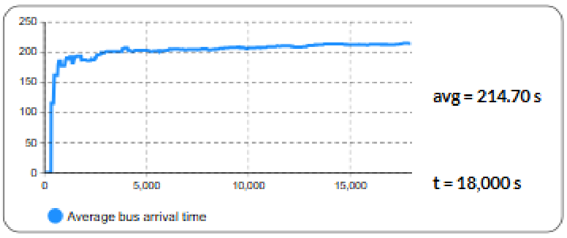
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

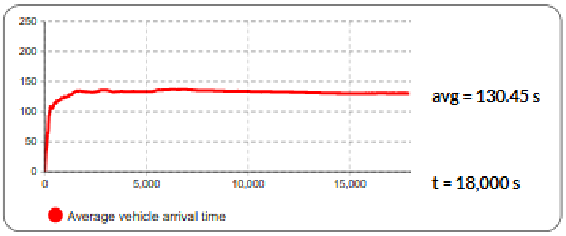



Average bus arrival time (seconds)

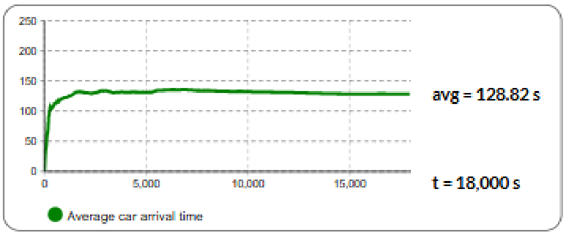



**Seed** fixed at a value of **2**:

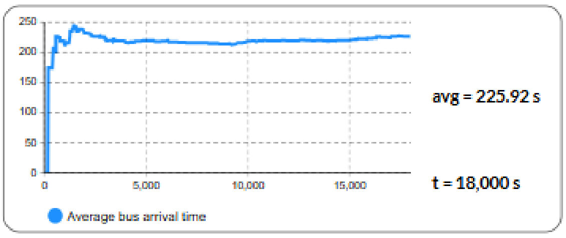
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

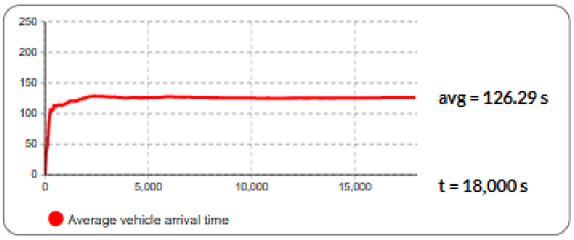



Average bus arrival time (seconds)

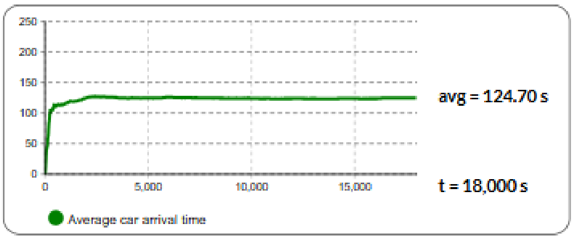



**Seed** fixed at a value of **3**:

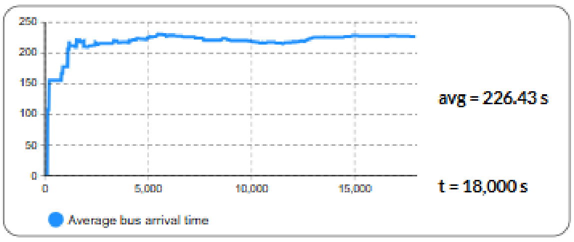
Average vehicle arrival time (seconds)




Average car arrival time (seconds)

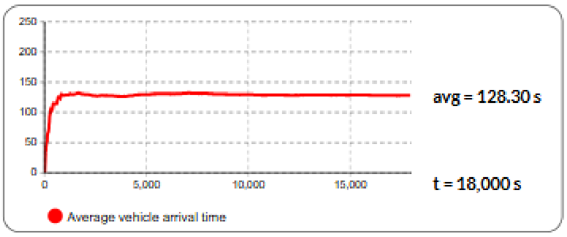



Average bus arrival time (seconds)

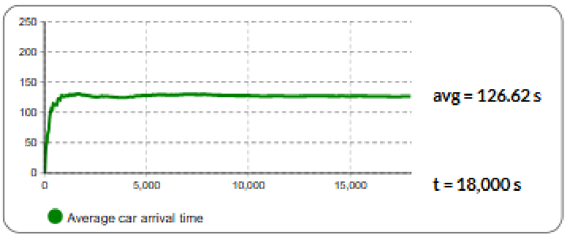



**Seed** fixed at a value of **4**:

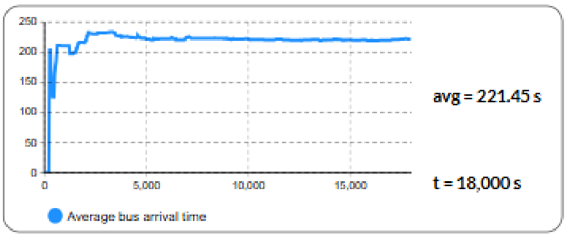
Average vehicle arrival time (seconds)




Average car arrival time (seconds)




Average bus arrival time (seconds)



### 4.4. Summary Tables

Below, we present three summary tables showing the average times for cars, buses, and a weighted average of both, for the three alternative systems:

| Weighted Average (Cars and Buses) | | | | |
|---|---|---|---|---|
| | | System | | |
| | | Baseline | Synchronized | Staggered |
| Seed | 1 | 123.00 | 128.11 | 126.95 |
| | 2 | 124.65 | 128.07 | 130.45 |
| | 3 | 120.59 | 127.28 | 126.29 |
| | 4 | 122.81 | 127.98 | 128.30 |

| Cars | | | | |
|---|---|---|---|---|
| | | System | | |
| | | Baseline | Synchronized | Staggered |
| Seed | 1 | 122.21 | 126.22 | 125.58 |
| | 2 | 123.75 | 126.04 | 128.82 |
| | 3 | 119.89 | 125.32 | 124.70 |
| | 4 | 122.16 | 126.38 | 126.62 |

| Buses | | | | |
|---|---|---|---|---|
| | | System | | |
| | | Baseline | Synchronized | Staggered |
| Seed | 1 | 165.30 | 238.50 | 214.70 |
| | 2 | 171.28 | 239.20 | 225.92 |
| | 3 | 161.10 | 238.45 | 226.43 |
| | 4 | 157.07 | 229.95 | 221.45 |

### 4.5. Analysis of Obtained Results

The results obtained from the simulations provide valuable information regarding the operation of the roundabout across the three proposed systems. We now proceed to present the main findings and conclusions derived from these results. To this end, values corresponding to the metrics of the data collected from the real system were estimated using confidence intervals, with a 95 % confidence level in the estimation.

To analyse a variable $X$, it is assumed that it follows a normal distribution; however, this variable will be estimated from multiple simulation runs to apply the Central Limit Theorem (CLT). To apply this theorem, a large quantity of data is required. It is estimated that 30 simulations are enough to proceed.

Since the variable's value is obtained from estimating an average, the parameters of the Normal distribution are:

$$\overline{X} \sim N\left(\overline{X}, \frac{S}{\sqrt{n}}\right)$$

where:

- $\bar{X}$ is the sample mean
- $S$ is the sample standard deviation
- $n$ is the sample size

The sample standard deviation is obtained using the following formula:

$$S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}}$$

To calculate confidence intervals, we proceed to standardize the variable of interest, transforming it into the standard normal random variable $z$.

If a random variable $X$ is normally distributed with mean $\mu$ and standard deviation $\sigma$, it follows that:

$$X \sim N(\mu, \sigma) \quad \Rightarrow \quad z = \frac{x-\mu}{\sigma}, \quad z \sim N(0,1)$$

Similarly, when working with the sample mean $\overline{X}$, which follows a normal distribution with mean $\mu$ and standard error $\frac{S}{\sqrt{n}}$, the standardization is expressed as:

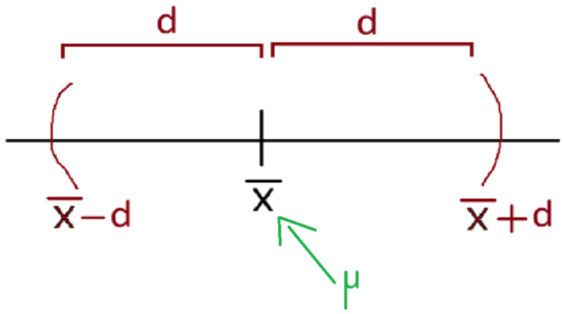
$$\overline{X} \sim N\left(\mu, \frac{S}{\sqrt{n}}\right) \quad \Rightarrow \quad z = \frac{\overline{X}-\mu}{S/\sqrt{n}}, \quad z \sim N(0,1)$$

This transformation allows using the critical values of the standard normal distribution to construct confidence intervals around the estimated mean.

Subsequently, as seen in the theoretical class development, the confidence interval for the mean is constructed based on the central probability of the standard normal distribution. This probability is given by:

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

where $z_{\alpha/2}$ represents the critical value of the standard normal variable corresponding to a significance level $\alpha$.




The previous figure graphically represents the confidence interval constructed around the sample mean $\overline{X}$, with a total width of $2d$. In this interval, the population parameter $\mu$ is contained, with a probability of $1-\alpha$, between the limits $\overline{X}-d$ and $\overline{X}+d$:

$$P(\overline{X}-d<\mu<\overline{X}+d)=1-\alpha$$

If a 95 % confidence interval is desired, then:

$$1-\alpha=0.95 \quad \Rightarrow \quad \alpha=0.05$$
$$\Rightarrow \quad \frac{\alpha}{2}=0.025$$

Thus, the critical value corresponding to that confidence level is:

$$z_{0.025}=1.96$$

Therefore, the interval's half-width (margin of error) is obtained via the following expression:

$$d=z_{\alpha/2}\cdot\frac{S}{\sqrt{n}}$$

We will analyse the average time spent in the system ($\overline{T}$) by vehicles across three distinct scenarios:

1. The **baseline system**, replicating real operation.
2. The **staggered system**, incorporating another set of traffic signals just prior to entering the roundabout, which will be staggered by 5 seconds relative to the previous signals.
3. The **synchronized system**, incorporating said set of traffic signals just prior to entering the roundabout, which will be synchronized with the previous signals.

**1. Baseline System:**

$\overline{T}=123.27$ seconds
$n=30$ simulations
$S=2.06$

Therefore, the value of $d$ is:

$d=0.74$ seconds

**Conclusion:** The Confidence Interval for $T$ is $(123.27-0.74;\ 123.27+0.74)$, that is:

$$(122.53;\ 124.01)$$

with 95 % confidence.

**2. Staggered System:**

$\overline{T}=128.28$ seconds
$n=30$ simulations
$S=1.69$

Therefore, the value of $d$ is:

$d=0.6048$ seconds

**Conclusion:** The Confidence Interval for $T$ is $(128.28-0.6048;\ 128.28+0.6048)$, that is:

$$(127.68;\ 128.88)$$

with 95 % confidence.

3. **Synchronized System:**

$\overline{T} = 129.818$ seconds
$n = 30$ simulations
$S = 1.75$

Therefore, the value of $d$ is:

$d = 0.63$ seconds

**Conclusion:** The Confidence Interval for $T$ is $(129.818 - 0.63;\ 129.818 + 0.63)$, that is:

$$(129.19;\ 130.45)$$

with 95 % confidence.

### 4.6. System Comparison

If there is no overlap of intervals, as in our case, the following statistic is used for comparison between systems:

$$Z_i = \overline{X}_{(1,i)} - \overline{X}_{(2,i)}$$

If $Z_i < 0$, the first system presents a lower value for the variable than the second system.
This means that if the variable corresponds to the *estimated average queue delay* ($\hat{d}$), the first system would be the best, since the average queue time is lower.
Conversely, if the variable represents the *estimated average server utilization* ($\hat{u}$), it is concluded that the second system is the most suitable, as it utilizes the server for a greater proportion of time.
Based on the results obtained, 95 % confidence intervals were constructed for the average time spent in the system ($\overline{T}$) in each analysed case:

- **Baseline System:**
  $IC = [122.53;\ 124.01]$

- **Staggered System:**
  $IC = [127.68;\ 128.88]$

- **Synchronized System:**
  $IC = [129.19;\ 130.45]$

It can be observed that there is no overlap between the confidence intervals, indicating that the differences between the systems are statistically significant at a 95 % confidence level.

As previously mentioned, for direct comparison between systems, the following expression is used:

$$Z_i = \overline{X}_{(1,i)} - \overline{X}_{(2,i)}$$

We reiterate that if $Z_i < 0$, the first system presents a lower value of the analysed variable regarding the second. In this case, given that the variable of interest is the average time spent in the system ($\overline{T}$), a lower value implies better performance (lower vehicle delay).

$$Z_{\text{Baseline-Staggered}} = 123.27 - 128.28 = -5.01 < 0$$

$$Z_{\text{Baseline-Synchronized}} = 123.27 - 129.818 = -6.548 < 0$$

$$Z_{\text{Staggered-Synchronized}} = 128.28 - 129.818 = -1.538 < 0$$

Therefore, the **Baseline System** presents the lowest average time spent in the system, followed by the **Staggered System**, and finally the **Synchronized System**, which proves to be the least efficient in terms of time.

If the analysed variable were *server utilization* ($\hat{u}$), where a higher value indicates better resource utilization, the interpretation would be reversed, and the system with the higher mean would be considered the most efficient.

## 5. Conclusions

Based on the 95 % confidence intervals calculated over the four seeds, the baseline system (replicating real operation without double signals) exhibits the lowest average time spent in the system for cars, buses, and the weighted average. However, this finding is insufficient to recommend it operationally: for certain seeds, the model enters critical congestion, queues grow without limit, and the simulation cannot proceed. In other words, the baseline scheme is fast when it works, but lacks robustness against specific pseudo-random outcomes depending on the seed.

In contrast, the two schemes featuring double traffic signals stabilize circulation: although they slightly increase the average time spent in the system, they prevent massive bottlenecks and ensure sustained flow throughout the entire simulation. Between both variants, the alternative where the signals on each access road are staggered by five seconds (with the signal furthest from the main roundabout turning green first) proves preferable: it reduces bus travel times (as it allows sufficient time to perform their stop, and upon resuming, find the roundabout entry signal green) and maintains average car times practically unchanged compared to the synchronized version.

Given these considerations, if system robustness (ensuring it never collapses) and improved public transport performance are prioritized, we recommend the double signal system staggered by five seconds, accepting a slight increase in the global average compared to the baseline system.

It rests with local authorities to weigh the social benefit of reducing public transport delay and decreasing the risk of severe bottlenecks against the slight increase in average time for private cars. Under this criterion, the double signal staggered by 5 seconds offers the best balance between fluidity and operational stability.

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*Mateo Seffino, +5493491405053, seffinomateo@gmail.com*